

EXP 10 -RECOMMENDATION SYSTEMS USING MATRIX FACTORIZATION (SVD OR NMF)

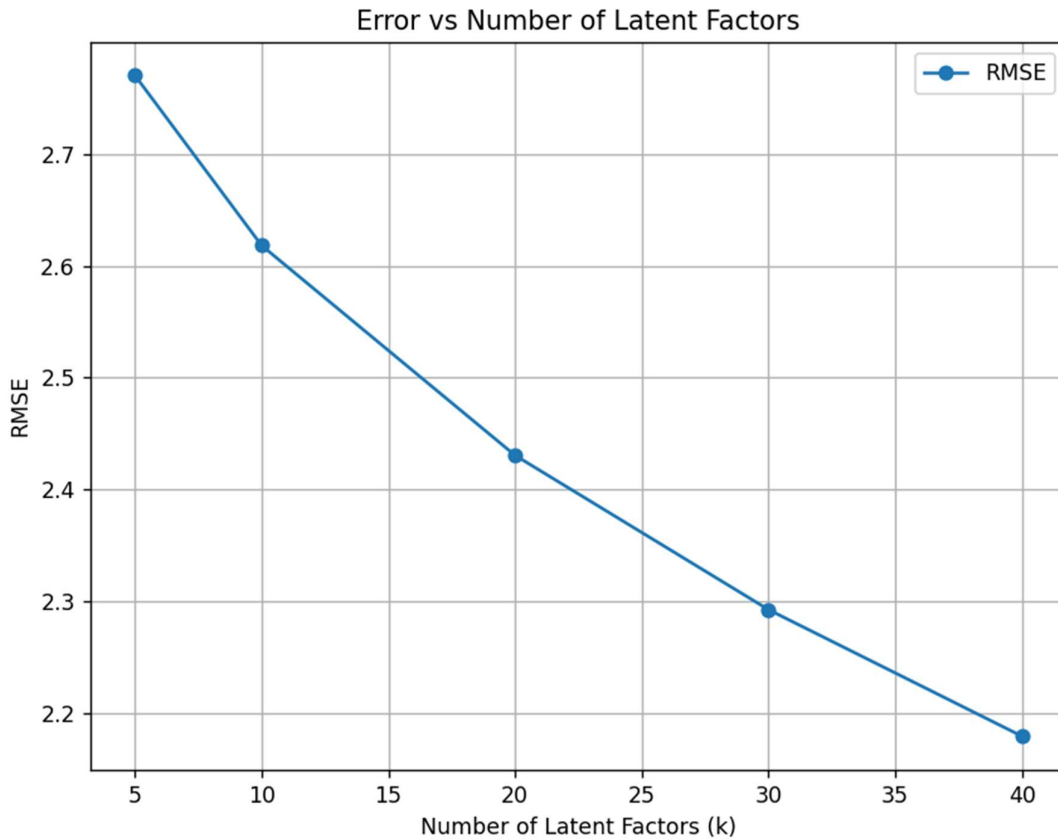
1.SCENARIO 1:MATRIX FACTORIZATION USING SVD

OUTPUT:

```
JANASREE 24BAD040
userId  movield  rating
0   1    31     2.5      Dangerous Minds (1995)
1   1   1029    3.0      Dumbo (1941)
2   1   1061    3.0      Sleepers (1996)
3   1   1129    2.0      Escape from New York (1981)
4   1   1172    4.0      Cinema Paradiso (Nuovo cinema Paradiso) (1989)
RMSE: 2.430676014267086
MAE: 2.0680108284142533

Top 10 Recommended Movies for User 1
title
Alien (1979)                                0.175545
Star Trek II: The Wrath of Khan (1982)      0.163278
Fantasia (1940)                             0.159034
Jaws (1975)                                 0.153054
Monty Python and the Holy Grail (1975)      0.146222
Lawrence of Arabia (1962)                   0.144503
African Queen, The (1951)                   0.142674
Monty Python's Life of Brian (1979)         0.142128
Who Framed Roger Rabbit? (1988)            0.136410
Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb (1964)  0.135617
Name: 1, dtype: float64
```

PLOT:



2.SCENARIO 2:MATRIX FACTORIZATION USING NMF

OUTPUT:

RMSE: 2.5298052187931748

Top 10 Recommended Movies for User 1

title	
Star Wars: Episode V - The Empire Strikes Back (1980)	0.156916
Alien (1979)	0.145342
Raiders of the Lost Ark (Indiana Jones and the Raiders of the Lost Ark) (1981)	0.143347
Star Wars: Episode VI - Return of the Jedi (1983)	0.135678
Terminator, The (1984)	0.133093
Star Wars: Episode IV - A New Hope (1977)	0.132668
Back to the Future (1985)	0.132175
Wizard of Oz, The (1939)	0.129496
Casablanca (1942)	0.124046
Butch Cassidy and the Sundance Kid (1969)	0.122326

Name: 1, dtype: float64

PLOT:

